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Combination Rules for van der Waals Force Constants

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A family of combination rules, from which van der Waals coefficients for interactions between unlike molecules may be estimated in terms of those for the corresponding like interactions and the static polarizabilities, is tested empirically and related to an approximate parameterization of the frequency-dependent polarizability function. The treatment includes two- and three-body dipole force constants and the higher-multipole terms of the second-order two-body dispersion energy. The rules give agreement within a few percent (about the same accuracy as the input data) when compared with accurate unlike coefficients for systems involving H, He, Ne, Ar, Kr, Xe, Li, Na, K, Rb, Cs, Hg, H_2, N_2, CH_4, $He(2^1S)$, $He(2^3S)$, He^+, Li^+, Na^+, K^+, Rb^+, Cs^+. Knowledge of n like coefficients and polarizabilities permits calculation of $n(n-1)/2$ unlike two-body and $n(n-1)(n+4)/6$ unlike three-body force constants.

Accurate prediction of intermolecular force constants for unlike partners from a knowledge of the corresponding constants for like interactions offers a great saving of effort for theoreticians and experimentalists alike. The geometric mean combination rule has been much used for such predictions, but with little justification and irregular success.[1] This paper documents a simple and excellent combination rule,

$$C_{ab}=\frac{2C_{aa}C_{bb}}{[(\alpha_b/\alpha_a)C_{aa}+(\alpha_a/\alpha_b)C_{bb}]}, \qquad (1)$$

where the C's are the familiar induced-dipole–induced-dipole van der Waals coefficients for pairwise interactions of atoms a and b and the α's are the static dipole polarizabilities. This rule apparently was first suggested by Moelwyn-Hughes[2] as early as 1957, although it has been ignored until recently.[3-5] The considerable body of accurate semiempirical C-constant calculations in the literature now makes possible a convincing test of Eq. (1). The C-constant calculations up to 1966 have been reviewed by Dalgarno,[6] and several new methods have recently been demonstrated by Langhoff and Karplus,[7] Gordon,[8] Davison,[9] Mahan,[10] and Weinhold.[11]

In this paper Eq. (1) is tested for a wide variety of systems including ground state atoms, molecules, and ions, and metastable atoms, and it is shown to be much superior to two other combination rules and to the three most commonly used approximate formulas for C constants. Some justification for the success of Eq. (1) is given in terms of the choice of parameters for a single-term approximation to the frequency-dependent polarizability function,[5] from which the van der Waals constant may be calculated. The same approach is employed to establish equally successful combination rules for the three-body dipole force constant. Finally, the method is generalized to obtain combination rules for all the higher multipole terms of the second-order two-body dispersion energy and, in the few cases for which accurate values are available, the results are again excellent.

APPROXIMATE FORMULAS

The three most widely used approximate formulas for C may be written in the form

$$C_{ab}=\tfrac{3}{2}\alpha_a\alpha_b\bar{\omega}_a\bar{\omega}_b/(\bar{\omega}_a+\bar{\omega}_b) \qquad (2a)$$

or, for like interactions,

$$C_{aa}=\tfrac{3}{4}\alpha_a^2\bar{\omega}_a. \qquad (2b)$$

Here α_a is the static dipole polarizability and $\bar{\omega}_a$ is some average energy for transitions from the initial state of system a. For the London (Lon) approximation[12] the choice of $\bar{\omega}_a$ is unspecified, two common choices being the ionization potential (I.P.) and the resonance or first transition energy ω_1 (Res). The latter choice clearly gives a lower bound to the C constant since any average energy for all transitions (such as that which the correct C requires) must be greater than the lowest contribution. For the Slater–Kirkwood (SK) approximation,[13]

$$\bar{\omega}_a=(N_a/\alpha_a)^{1/2}, \qquad (3)$$

where N_a is the number of electrons in the outer shell of a. For the Kirkwood–Müller (KM) approximation,[14]

$$\bar{\omega}_a=\tfrac{2}{3}\langle r^2\rangle_a/\alpha_a, \qquad (4)$$

where we have used

$$\langle r^2\rangle=\langle 0\mid\sum_{i=1}^{n}r_i^2\mid 0\rangle,$$

the value of r^2 summed over all (n) electrons and averaged over the wavefunction of the system, instead of the more usual parameter χ, the diamagnetic susceptibility, the two being related by a constant factor.[15] These three approximations have the advantage that they require a knowledge of some simple properties of a and b which may more easily be found than the C_{ab} constant itself, and the disadvantage that they are not highly accurate. It is reasonable to expect that better approximate values of C_{ab} could be obtained if the correct like coefficients C_{aa} and C_{bb} were used as parameters, as in Eq. (1). Indeed, the form of Eq. (2) is consistent with Eq. (1), for which the relevant choice of average energy is

$$\bar{\omega}_a = \tfrac{4}{3} C_{aa}/\alpha_a^2. \tag{5}$$

This will be referred to as the combination rule (CR) choice. A procedure algebraically equivalent to Eq. (5) has been used by Wilson.[16] Choosing an "empirical" value of N_a to reproduce an accurately determined C_{aa} with the SK-like formula, and using these N_a's in the SK unlike expression, he demonstrated implicitly that Eq. (1) predicts very good values of C_{ab} for rare-gas–rare-gas interactions. Tang[5] has confirmed these results and extended them to hydrogen-atom–rare-gas interactions.

It is convenient to define the parameter

$$G_{ab} = \alpha_a \alpha_b / C_{ab} \tag{6}$$

so that Eq. (1) may be written as

$$G_{ab} = \tfrac{1}{2}(G_{aa} + G_{bb}). \tag{7}$$

When $G_{aa} = G_{bb}$, Eq. (1) reduces to the geometric mean rule,

$$C_{ab} = (C_{aa} C_{bb})^{1/2}, \tag{8}$$

which only recently has been shown[5,11,17] to give an upper limit to C_{ab}. The occasional success of Eq. (8) now appears to arise from fortuitous application to cases with nearly equal $\bar{\omega}$ or G parameters; chemically similar species tend to have comparable $\bar{\omega}$ values. Another limiting case arises when $\alpha_a = \alpha_b$; Eq. (1) then becomes

$$C_{ab} = 2C_{aa}C_{bb}/(C_{aa} + C_{bb}). \tag{9}$$

Since the geometric mean of two different quantities is less than the arithmetic mean,

$$(C_{aa}C_{bb})^{1/2} < \tfrac{1}{2}(C_{aa} + C_{bb}),$$

the C_{ab} computed from Eq. (9) is always less than that computed from Eq. (8) and may in some cases be closer to the correct value.

EMPIRICAL TEST OF APPROXIMATIONS

The predictions of the Lon (I.P. and Res), SK, and KM formulas, and of the three combination rules of Eqs. (1), (8), and (9), have been compared with a set of published calculations (taken to be "true" values) of 153 unlike and 23 like C constants[18] for interactions of the following species: H, He, Ne, Ar, Kr, Xe, Li, Na, K, Rb, Cs, Hg, H_2, N_2, CH_4, He(2^1S), He(2^3S), He^+, Li^+, Na^+, K^+, Rb^+, Cs^+. The like C constants, static polarizabilities, average energies of Eq. (5), and G parameters are listed for reference in Table I. Values of the resonance energies, ionization potentials, N, and $\langle r^2 \rangle$ are tabulated elsewhere.[18] For each system (i), the fractional deviation of the approximate C constant

TABLE I. Accurate parameters for combination rules.[a]

System	C_{aa}	α_a	$\bar{\omega}_a$	G_{aa}
H	6.50	4.50	0.428	3.12
He	1.47	1.38	1.02	1.30
Ne	6.30	2.66	1.19	1.13
Ar	65.0	11.1	0.706	1.89
Kr	130.	16.7	0.619	2.15
Xe	270.	27.3	0.483	2.76
Li	1 390.	164.	0.0689	19.3
Na	1 580.	165.	0.0774	17.2
K	3 820.	293.	0.0593	22.5
Rb	4 600.	324.	0.0584	22.8
Cs	7 380.	432.	0.0527	25.3
Hg	240.	33.9	0.278	4.79
H_2	13.0	5.33	0.610	2.19
N_2	73.0	11.9	0.690	1.93
CH_4	150.	17.6	0.649	2.05
He(2^1S)	11 300.	802.	0.0234	57.0
He(2^3S)	3 290.	316.	0.0439	30.4
He^+	0.102	0.281	1.71	0.779
Li^+	0.101	0.211	3.03	0.440
Na^+	3.87	1.89	1.44	0.924
K^+	34.8	7.36	0.857	1.56
Rb^+	73.7	12.2	0.665	2.00
Cs^+	162.	18.4	0.640	2.08

[a] All quantities in atomic units; for C: $e^2a_0^5 = 0.957 \times 10^{-60}$ erg·cm^6; for α: $a_0^3 = 0.148$ Å^3; for $\bar{\omega}$ and $1/G$: $e^2/a_0 = 0.436 \times 10^{-10}$ erg. Values of C are from Ref. 6, except for Hg: W. C. Stwalley and H. L. Kramer, J. Chem. Phys. **49**, 5555 (1968); metastable He: G. A. Victor, A. Dalgarno, and A. J. Taylor, J. Phys. B **1**, 13 (1968); He^+: scaling C_{HH} by Z^{-6}, $Z=2$; and the alkali ions: R. A. Buckingham, Proc. Roy. Soc. (London) **A160**, 113 (1937). Values for α for H, H_2, N_2, and CH_4 are from J. O. Hirschfelder, C. F. Curtiss, and R. B. Bird, *Molecular Theory of Gases and Liquids* (Wiley, New York, 1954), pp. 946, 950; the rare gases: A. Dalgarno and A. E. Kingston, Proc. Roy. Soc. (London) **A259**, 424 (1960); the alkali metals: A. Dalgarno (private communication); Hg: W. C. Stwalley and H. L. Kramer, J. Chem. Phys. **49**, 5555 (1968); metastable He: G. A. Victor, A. Dalgarno, and A. J. Taylor, J. Phys. B **1**, 13 (1968); He^+: scaling α_H by Z^{-4}, $Z=2$; Rb^+: J. R. Tessman, A. H. Kahn, and W. Shockley, Phys. Rev. **92**, 890 (1953); other alkali ions: R. A. Buckingham, Proc. Roy. Soc. (London) **A160**, 94 (1937). Values of $\bar{\omega}$ and G are obtained from Eqs. (5) and (6).

(denoted by a prime) was computed,

$$D_i = [(C_{ab}' - C_{ab})/C_{ab}]_i,$$

and the root-mean-square (rms) deviation

$$R = [n^{-1} \sum_{i=1}^{n} D_i^2]^{1/2},$$

was evaluated for unlike coefficients in all seven approximations and for like coefficients in the first four approximations. These rms deviations are listed in Table II. Figure 1 presents histograms of the frequency of fractional deviations D_i for the unlike coefficients. An expanded histogram for Eq. (1) with a superposed Gaussian curve of standard deviation 2.5% is shown in Fig. 2.

The results for Eq. (1) are striking. The rms deviation of 3.25% is virtually tantamount to exact agreement

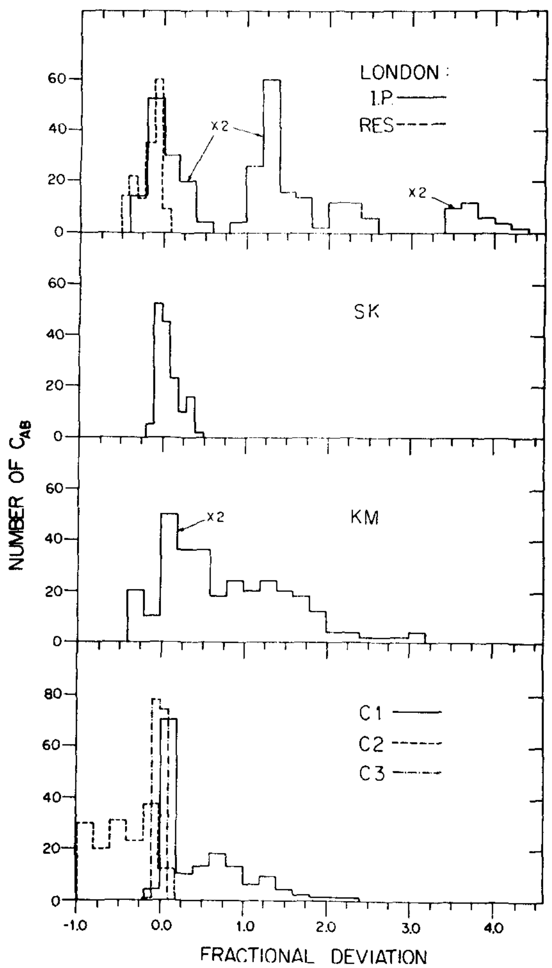

FIG. 1. Histograms showing distributions of fractional deviations from "true" values for various approximate C_{ab} constants (a unlike b): London, Eq. (2a) with lowest transition energy (Res) or ionization potential (I.P.); Slater–Kirkwood (SK), Eqs. (2a) and (3); Kirkwood–Müller (KM), Eqs. (2a) and (4); combination rules denoted by $C1$, $C2$, and $C3$, which refer to Eqs. (8), (9), and (1), respectively. The ordinates for the London (I.P.) and KM results have been doubled for ease of viewing.

TABLE II. Root-mean-square deviations of approximate C constants.[a]

Method	R (like)	R (unlike)
Lon—I.P.	149.	168.
Lon–Res	26.3	21.5
SK	15.9	15.4
KM	136.	110.
$C1$	···	73.5
$C2$	···	53.0
$C3$	···	3.25

[a] R (like) is the root-mean-square fractional deviation for 23 like–like interactions, expressed as percent; R (unlike), the same quantity for 153 like–unlike coefficients. $C1$, $C2$, and $C3$ refer to the combination rules set forth in Eqs. (8), (9), and (1), respectively.

since uncertainties in the "true" values of C and of α are often about 3% or larger.[6,19,20] There are several incidental observations of interest. For alkalilike species (including metastable He), the spectra of which are dominated by a single low-lying transition, use of the Lon (Res) formula gives a lower bound which almost precisely coincides with the true C coefficient. It is not surprising, therefore, that large positive deviations appear for interactions of alkalilike species in the Lon (I.P.) approximation. The Lon (I.P.) histogram in Fig. 1 shows three separate groups of deviations. The lowest group, which includes both positive and negative deviations, corresponds to all interactions not involving an alkalilike atom. The middle group contains all interactions of the alkalilike species other than $He(2^1S)$, while the highest group consists entirely of $He(2^1S)$

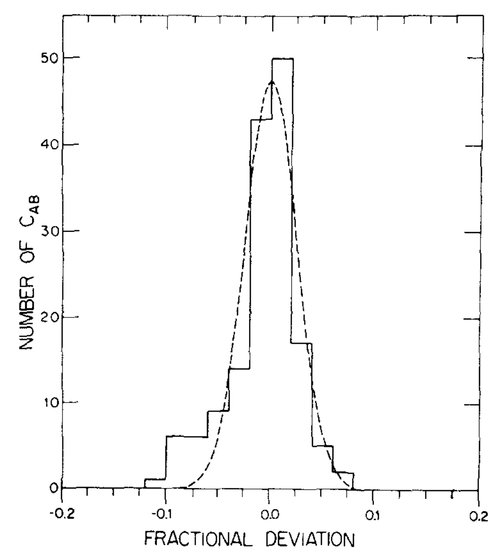

FIG. 2. Histogram distribution (—) of fractional deviations of C_{ab} constants ($a \neq b$) computed from the combination rule of Eq. (1), compared with a Gaussian distribution (- - -) of standard deviation 2.5%.

interactions. Likewise, for the neutral, closed-shell species considered here (inert gases, H_2, N_2, CH_4), the Lon (I.P.) formula invariably gives C constants that are too small. As noted by Wilson,[16] for the rare gases an improvement occurs in the SK approximation when the outer shell contains both s and p electrons if N is set equal to the number of p electrons only. Accordingly, we have used $N=6$ for the rare gases Ne through Xe, the alkali ions Na^+ through Cs^+, and N_2. The anomalously low empirical values of N that Wilson finds for Ne (3.76, 3.89) may be related to the fact that among the rare gases with outer shell configuration ns^2np^6, Ne alone, according to the tabulation of orbital radii by Waber and Cromer,[21] has the principal maximum in the radial distribution function for its outer s electrons occurring at a larger radius than that for its outer p electrons. Figure 1 shows that, with the exception noted for alkalilike species, the SK formula gives the best results of the first four approximations. This has been recognized by Barker and Leonard[22] and others.[19] More surprising is the observation that the distribution of deviations for the Lon (Res) formula has about the same width as the SK histogram, but is displaced downwards by 15%–20%. Thus, if our results are representative, one may expect to estimate C_{aa} or C_{ab} within 15%–20% either by direct use of the SK approximation or by increasing the value computed from the Lon (Res) approximation by 15%–20%.

THEORETICAL BASIS FOR APPROXIMATIONS

The second-order induced-dipole–induced-dipole dispersion force constant may be written[23]

$$C_{ab}=\frac{3}{\pi}\int_0^\infty \alpha_a(i\omega)\alpha_b(i\omega)d\omega, \tag{10}$$

where a and b label the interacting molecules and

$$\alpha(i\omega)=\sum_{m=1}^{M}\frac{f_m}{\omega_m{}^2+\omega^2} \tag{11}$$

is the frequency-dependent dipole polarizability evaluated for imaginary argument; f_m is the electric dipole oscillator strength for the transition from the initial (here, ground or metastable) state to the state m and ω_m is the transition energy. The exact $\alpha(i\omega)$ and C are given by sums for which the upper limit M is infinite and which include both discrete and continuum states, but for highly accurate calculations sums over a finite number of terms (sometimes as few as two) generally suffice.[6,23]

Any single-term representations of $\alpha_a(i\omega)$ and $\alpha_b(i\omega)$ will give coefficients C_{aa}, C_{bb}, and C_{ab} having the form of Eq. (2) and satisfying Eq. (1) identically. Thus, the Lon, SK, KM, and CR approximations of Eqs. (2)–(5) may be viewed as resulting from a particular choice of the two parameters which appear in single-term approximations to the correct $\alpha(i\omega)$ functions. All four approximations use the static polarizability $\alpha\equiv\alpha(0)$ as one parameter. This is clearly appropriate, for it ensures that the approximate one-term function, $\alpha'(i\omega)$, will coincide with the true function, $\alpha(i\omega)$, where the latter has its maximum value.

The general form of such a one-term function is

$$\alpha'(i\omega)=\alpha\bar{\omega}^2/(\bar{\omega}^2+\omega^2) \tag{12a}$$

or

$$\alpha'(i\omega)/\alpha=[1+(\omega/\bar{\omega})^2]^{-1} \tag{12b}$$

where $\bar{\omega}$ is the same energy parameter that appears in Eq. (2). The parameters α and $\bar{\omega}$ merely adjust the scales of ordinate and abscissa for the simple Lorentzian shape.

Tang[5] has proved the following theorem: If $\alpha'(i\omega)$ and $\alpha(i\omega)$ intersect for $\omega>0$, they intersect only once; furthermore, if ω_i is the point of intersection, then

$$\alpha'(i\omega)>\alpha(i\omega)\quad \text{for}\quad 0<\omega<\omega_i$$

and

$$\alpha'(i\omega)<\alpha(i\omega)\quad \text{for}\quad \omega_i<\omega<\infty.$$

This assigns the $\alpha'(i\omega)$ functions to three groups: (a) $\alpha'(i\omega)>\alpha(i\omega)$ for all $\omega>0$; (b) $\alpha'(i\omega)<\alpha(i\omega)$ for all $\omega>0$; and (c) $\alpha'(i\omega)$ intersects $\alpha(i\omega)$ once. Functions in group (a) will give upper bounds to C; those in group (b) will give lower bounds; and those in group (c) may give upper, lower, or no bounds.

It is useful to consider the behavior of the difference function

$$g(\omega)=\alpha'(i\omega)-\alpha(i\omega),$$

for limiting large and small values of ω. Substituting from Eqs. (11) and (12) and retaining only the leading terms, we find

$$\lim_{\omega\to 0} g(\omega)=\omega^2[S(-4)-\bar{\omega}^{-2}S(-2)] \tag{13}$$

and

$$\lim_{\omega\to\infty} g(\omega)=\omega^{-2}[\bar{\omega}^2S(-2)-S(0)], \tag{14}$$

where

$$S(k)=\sum_n f_n\omega_n{}^k,$$

and $\alpha=S(-2)$. Tang's group (a) corresponds to those functions for which (14) is nonnegative, or

$$\bar{\omega}\geq\omega_\infty=[S(0)/S(-2)]^{1/2}. \tag{15}$$

Similarly, group (b) contains those functions which make (13) nonpositive, or

$$\bar{\omega}\leq\omega_0=[S(-2)/S(-4)]^{1/2}. \tag{16}$$

Finally, the requirement for group (c) is that (13) be positive and (14) be negative:

$$\omega_0<\bar{\omega}<\omega_\infty. \tag{17}$$

The four approximations examined here differ solely in the way the average energy parameter $\bar{\omega}$ is assigned. The Lon approximation employs a guess based on partial knowledge of the spectrum of allowed transitions; the SK approximation requires $\alpha'(i\omega)$ and $\alpha(i\omega)$ to be equal for infinite argument; and the KM and CR approximations equate the areas under some power of $\alpha'(i\omega)$ and $\alpha(i\omega)$. These choices are compared in Fig. 3, which shows an accurate four-term representation[24] of $\alpha(i\omega)$ for He along with five one-term approximate functions and a corresponding plot of the fractional deviations $\Delta\alpha(i\omega)/\alpha(0)=[\alpha'(i\omega)-\alpha(i\omega)]/\alpha(0)$.

The Lon formula is essentially only a statement of the problem, since $\bar{\omega}$ is unspecified. No simple characteristic energy gives generally good C values. Since the right side of (16), which is a particular kind of average energy, must be greater than the resonance energy, $\alpha'(i\omega)$ for the Lon (Res) approximation will always be

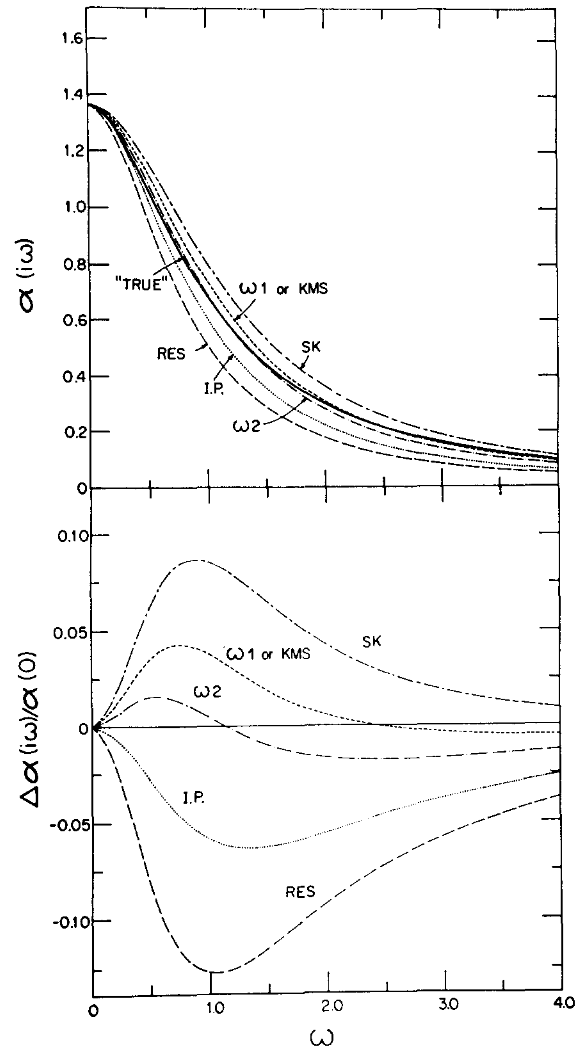

FIG. 3. Comparison of accurate four-term representation ("true") of the frequency-dependent polarizability of He with five one-term approximations (defined in text) which differ in the choice of the average energy parameters, $\bar{\omega}$. Upper panel: polarizability function versus frequency (atomic units); lower panel: deviations (approximation minus "true") expressed as a fraction of the static polarizability.

in group (b). In the case of He (Fig. 3) the Lon (I.P.) approximation also gives a one-term function in group (b).

The SK approximation, as first pointed out by Mavroyannis and Stephen,[25] corresponds to equating

$$\lim_{\omega\to\infty}\alpha'(i\omega)=\alpha\bar{\omega}^2/\omega^2$$

to

$$\lim_{\omega\to\infty}\alpha(i\omega)=(\Sigma_n f_n)/\omega^2.$$

This condition, together with the Thomas–Reiche–Kuhn sum rule,[26]

$$\sum_n f_n=S(0)=N_t,$$

where N_t is the total number of electrons in the atom, yields

$$\bar{\omega}=(N_t/\alpha)^{1/2}=[S(0)/S(-2)]^{1/2}. \qquad (18)$$

This form of the SK approximation, from Eq. (15), belongs to group (a) and gives an upper bound for C. However, as N_t increases, the upper limit diverges greatly from the true C value. The reason for the rather good results obtained with the usual form of the SK approximation, in which N_t is replaced by the number of outer shell electrons, N, is that essentially only the oscillator strengths for transitions involving outer electrons make important contributions to $\alpha(i\omega)$. For He, $N=N_t$, so that the SK curve in Fig. 3 deviates positively for all $\omega>0$.

For the KM and CR approximations, $\bar{\omega}$ is determined by a relation of the form

$$\int_0^\infty[\alpha'(i\omega)]^n d\omega=\int_0^\infty[\alpha(i\omega)]^n d\omega, \qquad (19)$$

giving an average energy we shall call ωn. Since the integrands in Eq. (19) weight smaller values of ω more heavily when n is larger,

$$\omega n>\omega n' \text{ for } n<n'. \qquad (20)$$

Any function $\alpha'(i\omega)$ satisfying Eq. (19) is in Tang's group (c): it must be somewhere greater and somewhere less than $\alpha(i\omega)$.

The KM approximation corresponds to $n=1$, or $\bar{\omega}=\omega 1$ with

$$\omega 1=\alpha^{-1}\sum_m f_m\omega_m^{-1}=S(-1)/S(-2). \qquad (21)$$

With the definition of oscillator strength,[23]

$$f_m=\tfrac{2}{3}\omega_m\,|\,\langle 0\,|\sum_{i=1}^{N_t}\mathbf{r}_i\,|\,m\rangle\,|^2,$$

the sum over states can be performed[26] and

$$S(-1)=\tfrac{2}{3}\langle 0\,|\,(\sum_i\mathbf{r}_i)\cdot(\sum_j\mathbf{r}_j)\,|\,0\rangle -\tfrac{2}{3}\,|\,\langle 0\,|\,(\sum_i\mathbf{r}_i)\,|\,0\rangle\,|^2.$$

The second term vanishes in the absence of a permanent dipole moment. The first term can be expanded as

$$\langle 0\,|\sum_i r_i^2\,|\,0\rangle+\langle 0\,|\sum_{i\neq j}\mathbf{r}_i\cdot\mathbf{r}_j\,|\,0\rangle,$$

and the usual KM approximation of Eq. (4) is obtained[27] by neglecting the part involving $\mathbf{r}_i\cdot\mathbf{r}_j$, which represents a negative electron correlation contribution. Figure 1 shows that Eq. (4) gives unlike C constants that are generally too large. Salem[27] proposed that the correlation term be retained (KMS approximation), which is equivalent to using $\omega 1$ of Eq. (21) rather than $\bar{\omega}$ of Eq. (4). When the complete $S(-1)$ is known, this procedure does yield a smaller C constant than the KM approximation, but the KMS result is still an upper bound to the true C constant, as shown in Eq. (22)

below. Although the $\alpha'(i\omega)$ for KMS will always intersect $\alpha(i\omega)$, that for KM will or will not intersect according to whether the $\bar{\omega}$ of Eq. (4) is less than or greater than that of Eq. (18).

Finally, the $\alpha'(i\omega)$ for CR is determined by Eq. (19) with $n=2$, or $\bar{\omega}=\omega 2$. With Tang's theorem, this ensures that the $\alpha'(i\omega)$ will exceed $\alpha(i\omega)$ for small ω and fall below it as $\omega\rightarrow\infty$. Tang[5] suggests that one-term polarizabilities with this property should predict fairly accurate C constants since the positive and negative deviations of $\alpha'(i\omega)$ from $\alpha(i\omega)$ tend to compensate when the integral of Eq. (10) over a pair of such functions is performed. Of the many possible choices of $\bar{\omega}$ which lead to intersecting polarizability functions, the choice $\omega 2$ made by the combination rule of Eq. (1), which ensures that $\alpha_a'(i\omega)$ and $\alpha_b'(i\omega)$ give the correct results for C_{aa} and C_{bb}, evidently is close to optimal for C_{ab}. Figure 3 shows for He that the $\omega 2$ function remains within a few percent of $\alpha(i\omega)$ in the important region of small ω.

A numerical test of results obtained with $n=1$, 2, 3 was made using 18 accurate fits to $\alpha(i\omega)$ which are available for the species of Table I other than the alkali ions.[18] These fits contain 2–11 terms; they substitute here for the exact functions. All the 153 possible unlike C constants were computed from Eq. (10) and compared with those given by Eq. (2a) applied in turn to each of $\omega 1$, $\omega 2$, and $\omega 3$. The rms deviations for these 153 interactions were 2.6% for $\omega 1$ (all deviations positive); 1.4% for $\omega 2$ (deviations mixed); and 2.0% for $\omega 3$ (nearly all deviations negative).

As a by-product, this discussion provides rigorous bounds on the true C constant. From Eqs. (5) and (20) we have

$$\omega 1>\omega 2=\tfrac{4}{3}C/\alpha^2.$$

Since both $\omega 1$ and $\omega 2$ give group (c) functions, Eq. (17) implies that

$$\tfrac{3}{4}\alpha^2\omega_0<C<\tfrac{3}{4}\alpha^2\omega 1<\tfrac{3}{4}\alpha^2\omega_\infty. \qquad (22)$$

COMBINATION RULES FOR THREE-BODY COEFFICIENTS

The leading second-order term in the dispersion energy of three spherical atoms is just the sum of the three two-body interactions. A nonadditive three-body dipole–dipole–dipole term arises in third order,[6,24] the coefficient of which may be written as an integral over three dipole polarizability functions,[23]

$$\gamma_{abc}=\frac{3}{\pi}\int_0^\infty \alpha_a(i\omega)\alpha_b(i\omega)\alpha_c(i\omega)d\omega. \qquad (23)$$

The three-body analog of Eq. (2a) is obtained by substituting the single-term approximate polarizability functions of Eq. (12) and integrating, with the result

$$\gamma_{abc}=\frac{3}{2}\frac{\alpha_a\alpha_b\alpha_c\bar{\omega}_a\bar{\omega}_b\bar{\omega}_c(\bar{\omega}_a+\bar{\omega}_b+\bar{\omega}_c)}{(\bar{\omega}_a+\bar{\omega}_b)(\bar{\omega}_b+\bar{\omega}_c)(\bar{\omega}_c+\bar{\omega}_a)}, \qquad (24a)$$

or, for like interactions,

$$\gamma_{aaa}=\tfrac{9}{16}\alpha_a{}^3\bar{\omega}_a. \qquad (24b)$$

Tang[5] has suggested that average energies for Eqs. (24) may be chosen in the same ways as for approximate two-body coefficients. Thus, analogs to the Lon, SK, and KM approximations are obtained by substituting the appropriate $\bar{\omega}$'s. A combination rule analogous to Eq. (1) is obtained by choosing $\bar{\omega}$ to satisfy (19) with $n=3$; from Eq. (23) with $a=b=c$, this gives $\bar{\omega}=\omega 3$ with

$$\omega 3=(16/9)\gamma_{aaa}/\alpha_a{}^3. \qquad (25)$$

In terms of the parameter

$$G_{abc}=\tfrac{3}{4}\alpha_a\alpha_b\alpha_c/\gamma_{abc}, \qquad (26)$$

the three-body combination rule resulting from $\bar{\omega}=\omega 3$ in Eq. (24a) is given by

$$G_{abc}=\frac{3}{8}\frac{(G_{aaa}+G_{bbb})(G_{bbb}+G_{ccc})(G_{ccc}+G_{aaa})}{G_{aaa}G_{bbb}+G_{bbb}G_{ccc}+G_{ccc}G_{aaa}}. \qquad (27)$$

However, a more useful rule can be obtained in terms of two-body like coefficients by employing $\bar{\omega}_a=\omega 2=\tfrac{3}{4}G_{aa}$ in Eq. (24a), which gives

$$G_{abc}=\frac{3}{8}\frac{(G_{aa}+G_{bb})(G_{bb}+G_{cc})(G_{cc}+G_{aa})}{G_{aa}G_{bb}+G_{bb}G_{cc}+G_{cc}G_{aa}}. \qquad (28)$$

These two rules are stated in a different form by Tang.[5] An approximate expression for γ_{abc} in terms of the unlike C_{ab} constants has also been derived by Midzuno and Kihara.[28] With Eqs. (1), (6), and (26), their relation becomes the same as Eq. (28).

As in the two-body case, Eq. (20) leads incidentally to some bounds for γ. Thus, $\omega 2>\omega 3$ implies $G_{aa}<G_{aaa}$, or

$$\gamma_{aaa}<\tfrac{3}{4}\alpha_aC_{aa}. \qquad (29)$$

This inequality was derived by Tang and shown to give a tight upper bound to γ_{aaa} for H and the rare gases[5]; written as an equality, it is the diagonal case of the Midzuno–Kihara approximation.[28] A lower bound and a looser upper bound result from replacing $\bar{\omega}$ in Eq. (17) by $\omega 3$ from Eq. (25), which yields

$$(9/16)\alpha^3\omega_0<\gamma<(9/16)\alpha^3\omega_\infty. \qquad (30)$$

A two-part numerical test of the combination rules of Eq. (27) and (28) was made. The first (I) made use of a set of ("true") calculated values of γ_{abc} for 8 like and 97 unlike interactions published by Bell and Kingston[29] and Dalgarno, Morrison, and Pengelly[30] for the species H, He, Ne, Ar, Kr, Xe, H_2, and N_2. These values were used together with the polarizabilities and diagonal C constants listed in Table I for these species. The second part (II) employed the 18 dipole polarizability fits cited previously[18] to compute all of the 18 possible diagonal and 1122 off-diagonal γ_{abc}, as well as the 18 static polarizabilities and diagonal C constants. The second calculation, in addition to including many more cases than the first, should pro-

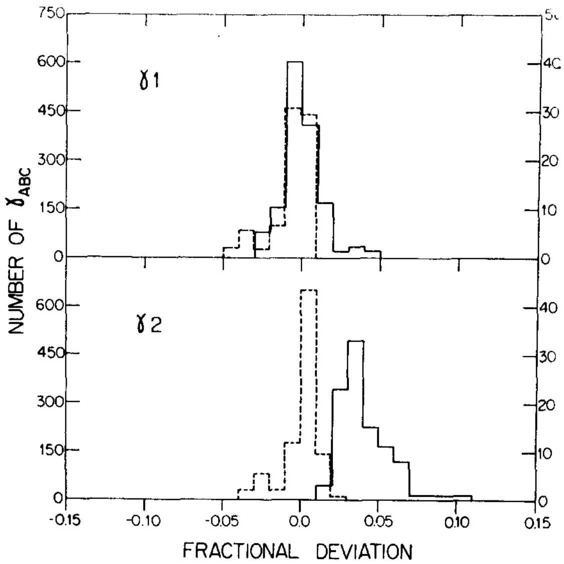

FIG. 4. Histogram distributions of fractional deviations from "true" values (see text) of approximate three-body coefficients γ_{abc}. Two distinct sets of "true" values were employed: One set (—, with right-hand ordinate scale) comprises accurate calculations (Ref. 29 and 30) of γ_{abc} for 8 like and 97 unlike interactions among the species H, He, Ne, Ar, Kr, Xe, H_2 and N_2, along with the C_{aa} constants and polarizabilities of Table I; the other set (---, with left-hand ordinate scale) uses C_{aa} and γ_{aaa} constants and polarizabilities for 18 systems as well as all 1122 possible unlike γ_{abc} computed from the dipole polarizability fits for these systems (Ref. 18). $\gamma 1$ and $\gamma 2$ refer to the three-body combination rules given in Eqs. (27) and (28), respectively.

vide a more rigorous test of the combination rules since the numbers used are internally consistent. Figure 4 presents histograms showing the frequencies of fractional deviations $D=(\gamma_{abc}'-\gamma_{abc})/\gamma_{abc}$, of the approximate coefficient (primed) from the "true" value (for I or II). The upper part of the figure gives the results for Eq. (27), labeled $\gamma 1$, the lower part the results for Eq. (28), labeled $\gamma 2$. The rms deviations for $\gamma 1$ are 1.13% for I and 1.35% for II; for $\gamma 2$, 4.36% for I and 1.13% for II. Although the deviations for II are predominantly positive for Eq. (28), that relation does not give a rigorous upper bound for unlike coefficients. On the other hand, it is readily shown that since $\omega 2>\omega 3$, the approximation for γ_{abc} given by Eq. (28) is an upper bound to the coefficient given by Eq. (27), which may in turn deviate either positively or negatively from the correct value. The proof of this consists merely in showing that in Eq. (24a) the derivative of γ_{abc} with respect to any of the average energies is positive.

HIGHER MULTIPOLE TWO-BODY COEFFICIENTS

The long-range second-order dispersion energy between two spherical systems is given by[23]

$$V_{ab}(R)=-\sum_{l_a=1}^{\infty}\sum_{l_b=1}^{\infty}\mathcal{E}_{ab}(l_a,l_b)R^{-2(l_a+l_b+1)},\qquad(31)$$

where

$$\mathcal{E}_{ab}(l_a,l_b)=\frac{(2l_a+2l_b)!}{4(2l_a)!(2l_b)!}\left(\frac{2}{\pi}\right)\int_0^\infty \alpha_{l_a}(i\omega)\alpha_{l_b}(i\omega)d\omega\qquad(32)$$

is the coefficient of the interaction of the induced multipole of order 2^{l_a} on system a with that of order 2^{l_b} on b, and $\alpha_{l_a}(i\omega)$ is the dynamic 2^{l_a}-pole polarizability at frequency $i\omega$, given by an expression of the same form as Eq. (11), with the 2^{l_a}-pole oscillator strength defined by

$$f_m^{(l_a)}=2\omega_m\left|\langle 0\left|\sum_i r_i^{l_a}P_{l_a}(\cos\theta_i)\right|n\rangle\right|^2.$$

Here $P_{l_a}(\cos\theta_i)$ is the Legendre polynomial of order l_a and the subscript i labels the electrons. The familiar coefficients of the R^{-6}, R^{-8}, and R^{-10} terms are given in terms of the $\mathcal{E}_{ab}(l_a,l_b)$ coefficients by

$$C=\mathcal{E}_{ab}(1,1),\qquad(33a)$$

$$C_8=\mathcal{E}_{ab}(1,2)+\mathcal{E}_{ab}(2,1),\qquad(33b)$$

and

$$C_{10}=\mathcal{E}_{ab}(1,3)+\mathcal{E}_{ab}(2,2)+\mathcal{E}_{ab}(3,1).\qquad(33c)$$

In analogy to the dipole case, one-term approximations of the same form as Eq. (12) can be constructed for the higher-multipole polarizabilities. Integration of Eq. (32) with these one-term functions then provides a generalization of Eq. (2a), given by

$$\mathcal{E}_{ab}(l_a,l_b)=\frac{(2l_a+2l_b)!}{4(2l_a)!(2l_b)!}\,\frac{\alpha_{l_a}\alpha_{l_b}\bar{\omega}_{l_a}\bar{\omega}_{l_b}}{\bar{\omega}_{l_a}+\bar{\omega}_{l_b}}.\qquad(34)$$

Application of Eq. (19) with $n=2$ specifies the mean energies,

$$\bar{\omega}_{l_a}=8[(2l_a)!]^2\mathcal{E}_{aa}(l_a,l_a)/[(4l_a)!\alpha_{l_a}^2].\qquad(35)$$

With the definition

$$G_{ab}(l_a,l_b)=\alpha_{l_a}\alpha_{l_b}/\mathcal{E}_{ab}(l_a,l_b),\qquad(36)$$

these relations give the generalized two-body combination rule as

$$G_{ab}(l_a,l_b)=\tfrac{1}{2}[F(l_a,l_b)G_{aa}(l_a,l_a)+F(l_b,l_a)G_{bb}(l_b,l_b)]\qquad(37)$$

with

$$F(l,l')=[(4l)!(2l')!]/[(2l)!(2l+2l')!].$$

A few special cases may be noted. If $l_a=l_b=l$, the rule

TABLE III. Test of approximation for $\mathcal{E}(1,2)$ coefficient.[a]

Interaction	α_2	$\mathcal{E}(1,1)$	$\mathcal{E}(2,2)$	$\mathcal{E}(1,2)$	$\mathcal{E}'(1,2)$
H–H	15.0	6.5	1135.	62.2	62.2
He–He	2.35	1.47	63.5	7.05	7.02

[a] The static quadrupole polarizabilities α_2 are taken from A. Dalgarno, Advan. Phys. 11, 281 (1962); the "true" values, $\mathcal{E}(1,1)$, $\mathcal{E}(2,2)$, and $\mathcal{E}(1,2)$, are from J. O. Hirschfelder and W. J. Meath, Advan. Chem. Phys. 12, 3 (1967), for H, and from Ref. 6 for He. The approximate value $\mathcal{E}'(1,2)$ is obtained from Eq. (40).

reduces to

$$G_{ab}(l,l)=\tfrac{1}{2}[G_{aa}(l,l)+G_{bb}(l,l)]. \tag{38}$$

For $l=1$ this is Eq. (1) or (7). If $a=b$ but $l_a\neq l_b$, the rule is

$$G_{aa}(l_1,l_2)=\tfrac{1}{2}[F(l_1,l_2)G_{aa}(l_1,l_1)+F(l_2,l_1)G_{aa}(l_2,l_2)], \tag{39}$$

where subscripts 1 and 2 are used in place of a and b to avoid confusion. The simplest case is $l_1=1$ and $l_2=2$, for which

$$G_{aa}(1,2)=(1/5)G_{aa}(1,1)+(7/3)G_{aa}(2,2). \tag{40}$$

A recursion relation is obtained by solving Eq. (39) for $G_{aa}(l_2,l_2)$, changing a to b and l_2 to l_b, and substituting for $G_{bb}(l_b,l_b)$ in Eq. (37), which gives

$$G_{ab}(l_1,l_2)=G_{bb}(l_1,l_2)+\tfrac{1}{2}F(l_1,l_2)[G_{aa}(l_1,l_1)-G_{bb}(l_1,l_1)]. \tag{41}$$

This relation is most likely to be useful for $l_2>l_1$, as it then requires minimal knowledge of higher-multipole coefficients. For $l_1=1$ and $l_2=2$, it becomes

$$G_{ab}(1,2)=G_{bb}(1,2)+\tfrac{1}{5}[G_{aa}(1,1)-G_{bb}(1,1)]. \tag{42}$$

Since accurate higher-order coefficients are known for only a few systems, a thorough empirical test cannot yet be made for these more general relations. Nevertheless, they can be expected to have accuracy comparable to Eq. (1), and this is supported by the few examples for which numbers are available. Table III gives estimates of $\mathcal{E}_{aa}(1,2)$ for H and He obtained from Eq. (40). The agreement with theoretical values is excellent. A test of Eq. (42) was made for 18 systems in which b is a hydrogen atom and a is in turn each of the species, excepting the alkali ions, for which Eq. (1) was tested earlier. For this purpose the 18 dipole polarizability fits[18] and a quadrupole polarizability fit for the H atom[31] were used. The "true" values of $\mathcal{E}_{aa}(1,1)$ and $\mathcal{E}_{ab}(1,2)$ for 18 species a, and $b=$H, were computed from Eq. (32). The comparison of these $\mathcal{E}_{ab}(1,2)$ coefficients with those found by substituting "true" values on the right side of Eq. (42) is presented in Table IV. (The trivial case $a=$H is omitted.) Agreement is again excellent, the rms error being 1.4%. It should be noted, however, that this calculation provides only part of the C_8 constants for these interactions, according to Eq. (33b). The other contribution, $\mathcal{E}_{ab}(2,1)$, can be derived from a combination rule only if the quadrupole polarizability of the a species and an accurate value for some coefficient involving an induced quadrupole on a ($l_a=2$) are known.

TABLE IV. Test of approximation for estimating $\mathcal{E}_{ab}(1,2)$ for H-atom induced quadrupole interactions.[a]

Specie	$\mathcal{E}_{ab}(1,2)$	$\mathcal{E}_{ab}'(1,2)$
He	28.55	28.53
Ne	58.82	59.91
Ar	207.3	210.6
Kr	291.2	294.0
Xe	405.9	407.8
Li	576.5	570.1
Na	635.3	636.6
K	875.7	877.3
Rb	939.6	940.8
Cs	1105.	1106.
Hg	357.8	357.9
H_2	84.44	84.53
N_2	205.4	205.5
CH_4	294.2	294.4
He(2^1S)	1031.	997.7
He(2^3S)	734.4	709.7
He^+	6.89	6.83

[a] "True" coefficients $\mathcal{E}_{ab}(1,2)$ (with $b=$H) were computed from Eq. (32) using the polarizability fits tabulated in Ref. 18, as were "true" values of the quantities required for the right side of Eq. (42). The latter values were used to obtain approximate coefficients $\mathcal{E}_{ab}'(1,2)$ from Eq. (42).

DISCUSSION

The family of combination rules provided by Eqs. (1), (28), and (37) appears to be accurate within ~3% or better, which is much superior to present-day experimental results[1,32] and compares favorably with the best theoretical calculations.[6] No systematic trends in the accuracy of Eq. (1) are discernible, despite a wide variation in the relative size of the G_{aa} and G_{bb} parameters. Although the combination rules have been well tested only for atom–atom interactions, the general character of dispersion forces suggests the rules will be satisfactory for molecules also. The results obtained for interactions involving H_2, N_2, and CH_4 indeed proved to be as good as for atom–atom cases. For practical purposes, these rules reduce the problem of evaluating dispersion forces to that of finding the static polarizabilities and the coefficients of like interactions only. Knowledge of the polarizability and the $G_{aa}(1,1)$ parameter for n species enables C_{ab} to be predicted for $\frac{1}{2}n(n-1)$ unlike interactions, γ_{aaa} for n interactions, γ_{aab} for $n(n-1)$, and γ_{abc} for $\frac{1}{6}n(n-1)(n-2)$ interactions. Knowledge of $G_{aa}(1,2)$ for n' of these species ($n'<n$) enables $G_{ab}(1,2)$ to be predicted for $n'(n-1)$ interactions. Thus, the $n=23$ G_{aa} parameters given in Table I are equivalent to 253 C_{ab} coefficients and 2277 γ_{abc} coefficients. Whenever an accurate value of either G_{aa} or G_{ab} is obtained for any species a outside the presently known set (assumed to include b), the rules may be used to generate the coefficients for interaction of that species with all the others. There is no reason, furthermore, why the rules may not be used to estimate the static polarizabilities in cases where the remaining necessary quantities are all known.

The amplification offered by the combination rules should encourage a more systematic development of dispersion force data. Experimental results may soon show marked improvement. Molecular beam scattering, which is the most direct probe of dispersion forces, is now capable of providing accurate ratios of C_{ab} constants for many systems, since computerized normalization procedures[33] and mass spectrometric detection[34] are coming into general use. According to Eq. (1), for a pair of target gases (a and a') and a given beam species (b), we have

$$C_{a'b}/C_{ab}=[\alpha_{a'}(G_{aa}+G_{bb})/\alpha_a(G_{a'a'}+G_{bb})].$$

A greatly enlarged table of G_{aa} parameters thus can be systematically built up and cross checked by accurate measurements of such ratios, as the target gas is readily varied and polarizability values of good quality are available from refractive index data for many molecules.[35] However, aside from limited accuracy, the large number of $C_{a'b}/C_{ab}$ ratios already determined for scattering of alkali atom beams[1,32] is useless for this purpose; these ratios essentially reduce to $\alpha_{a'}/\alpha_a$ because G_{bb} is so large (Table I) for an alkali atom.

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[1] See, for example, H. Pauly and J. P. Toennies, Advan. At. Mol. Phys. **1**, 195 (1965).
[2] E. A. Moelwyn-Hughes, *Physical Chemistry* (Pergamon, New York, 1957), p. 332.
[3] D. M. Young and A. D. Crowell, *Physical Adsorption of Gases* (Butterworths, London, 1962), p. 10.
[4] A. D. Crowell, J. Chem. Phys. **49**, 3324 (1968).
[5] K. T. Tang, Phys. Rev. **177**, 108 (1969).
[6] A. Dalgarno, Advan. Chem. Phys. **12**, 143 (1967).
[7] P. W. Langhoff and M. Karplus, Phys. Rev. Letters **19**, 1461 (1967).
[8] R. G. Gordon, J. Chem. Phys. **48**, 3929 (1968).
[9] W. D. Davison, J. Phys. B **1**, 597 (1968).
[10] G. D. Mahan, J. Chem. Phys. **48**, 950 (1968).
[11] F. Weinhold, J. Phys. B **2**, 517 (1969).
[12] F. London, Trans. Faraday Soc. **33**, 8 (1937).
[13] J. C. Slater and J. G. Kirkwood, Phys. Rev. **37**, 682 (1931).
[14] J. G. Kirkwood, Z. Physik **33**, 57 (1932); A Müller, Proc. Roy. Soc. (London) **A154**, 624 (1936).
[15] J. H. Van Vleck, *The Theory of Electric and Magnetic Susceptibilities* (Oxford U. P., London, 1932), p. 91.
[16] J. N. Wilson, J. Chem. Phys **43**, 2564 (1965).
[17] K. T. Tang, J Chem Phys **49**, 4727 (1968).
[18] See H. L. Kramer, Ph.D. thesis, Harvard University, 1969, for parameters and references.
[19] E. A. Mason and L. Monchick, Advan. Chem. Phys. **12**, 329 (1967).
[20] Since Eq. (8) gives a rigorous upper bound to the true C constant, the fact that four fractional deviations for the C1 approximation are negative in Fig. 1 must certainly be due to inconsistency between the like and unlike constants for those cases. A similar inconsistency must account for the fact that nine values of D_i for the Lon (Res) formula, a lower bound to C, are positive.
[21] J. T. Waber and D. T. Cromer, J. Chem. Phys. **42**, 4116 (1965).
[22] J. A. Barker and P. J. Leonard, Phys. Letters **13**, 127 (1964).
[23] A. Dalgarno and W. D. Davison, Advan. At. Mol. Phys. **2**, 1 (1966).
[24] A. Dalgarno and G. A. Victor, Mol. Phys. **10**, 333 (1966).
[25] C. Mavroyannis and M. J. Stephen, Mol. Phys. **5**, 629 (1962).
[26] J. O. Hirschfelder, W. B. Brown, and S. T. Epstein, Advan. Quantum Chem. **1**, 255 (1964).
[27] L. Salem, Mol. Phys. **3**, 441 (1960).
[28] Y. Midzuno and T. Kihara, J. Phys. Soc. Japan **11**, 1045 (1956).
[29] R. J. Bell and A. E. Kingston, Proc. Phys. Soc. (London) **88**, 901 (1966).
[30] A. Dalgarno, I. H. Morrison, and R. M. Pengelly, Intern. J. Quantum Chem. **1**, 161 (1967).
[31] A. Dalgarno (private communication).
[32] R. B. Bernstein and J. T. Muckerman, Advan. Chem. Phys. **12**, 389 (1967).
[33] R. Düren, A. Frick, and Ch. Schlier, Z. Physik (to be published).
[34] See, for example, D. H. Winicur, A. L. Moursund, W. R. Devereaux, L. R. Martin, and A. Kuppermann, J. Chem. Phys. **52**, 3299 (1970); N. C. Blais and J. B. Cross, *ibid.* 3580 (1970), and references cited therein.
[35] See, for example, W. Brandt, J. Chem. Phys. **24**, 501 (1956); E. W. Rothe and R. B. Bernstein, *ibid.* **31**, 1619 (1959), and references cited therein.

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ν_4 Band of $^{13}CH_4$

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Absorption spectra of methane enriched to 52% in the ^{13}C isotope were obtained in the 1300-cm^{-1} region with a 0.45-cm^{-1} resolution. The theoretical method of Hecht was used to determine the rotational and splitting parameters for the ν_4 band of $^{13}CH_4$. A line-by-line calculated spectrum, using the calculated line positions, is compared to the observed spectrum.

INTRODUCTION

In a series of papers Jahn *et al.*[1–4] provided the first quantitative basis for explaining the line splittings observed in the infrared spectrum of methane. A more extensive treatment has been given by Herranz,[5] which is often convenient for approximate calculations. A more refined and elegant treatment by Hecht[6] was published at approximately the same time as the work of Herranz. This treatment was used by Hecht[7] to analyze high-resolution spectra of the ν_3 band of $^{12}CH_4$ obtained by Plyler, Tidwell, and Blaine.[8] The ν_3 band of $^{13}CH_4$ was later studied by McDowell,[9] who used the